Grids Gold Customization

Development Kickoff Brief for the Quantro Team

1. Objective

The development team shall customize the existing **Quantro multi-tenant system** to support the operational requirements of jewelry and gold businesses.

The team must **extend the current Quantro architecture**, not create a separate disconnected application. Existing modules such as tenants, companies, branches, users, permissions, items, inventory, purchasing, POS, invoicing, customers, payments, and reports shall remain the foundation of the system.

The first delivery must focus on a commercially usable **Jewelry Retail Core**, including:

- Jewelry item management.
- Gold and metal attributes.
- Karat and purity management.
- Weight-based inventory.
- Daily gold rates.
- Stones and certificates.
- Making charges and wastage.
- Jewelry price calculation.
- Gold-aware POS and invoicing.
- Tenant, company, branch, and location isolation.

This direction is aligned with the SRS Phase 1 scope, which prioritizes identity, branches, item master, inventory, purchasing, POS, customers, pricing, invoicing, finance, and reports before manufacturing and enterprise extensions.

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2. Core Architectural Rule

Quantro is already a tenant-based system. All jewelry customizations must therefore remain fully compatible with the existing multi-tenant model.

Every business record must belong to the correct:

- Tenant or company.
- Branch.
- Inventory location, where applicable.
- Authorized user and permission scope.

No company shall be able to access another company's items, gold rates, customers, invoices, stock, prices, or reports.

The SRS requires isolated operational data per company, multiple branches per company, operational locations such as showrooms, warehouses, safes, trays, workshops, and transit locations, as well as configurable metal types, karats, and stone types.

Required Tenant Controls

The team must ensure that:

1. All new tables include `company_id` or the existing Quantro tenant identifier.
2. Branch-sensitive records include `branch_id`.
3. Inventory records include `location_id` when required.
4. All API queries are automatically tenant-scoped.
5. Item codes are unique inside each company, not globally across all tenants.
6. Gold rates may be configured separately by company.
7. Optional branch-specific gold rates and pricing policies are supported.
8. Unauthorized users cannot view safe, vault, or restricted-location inventory.
9. Price changes, rate changes, adjustments, voids, and overrides are auditable.

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3. Jewelry Item Master Customization

The existing Quantro item or product module must be extended into a **Jewelry Item Master**.

The item module must support ordinary products and jewelry-specific products without breaking existing Quantro functionality.

3.1 Item Types

At minimum, support:

- Serialized jewelry piece.
- Weighted gold item.
- Style-based item.
- Jewelry set or bundle.
- Service item.
- Non-stock item.

Examples include:

- Ring.
- Necklace.
- Bracelet.
- Earrings.
- Gold bar.
- Gold coin.
- Diamond item.
- Silver item.
- Repair service.
- Making or customization service.

3.2 Required General Item Fields

Retain the existing Quantro item information, including:

- Item ID.
- Item code or SKU.
- Product name.
- Category and subcategory.
- Collection.
- Brand.
- Style or design code.
- Description.
- Barcode.
- QR code.
- Serial number.
- Images and attachments.
- Active or archived status.

3.3 Required Gold and Metal Fields

Add the following jewelry-specific fields:

Field	Purpose
`metal_type`	Gold, silver, platinum, or another configured metal
`karat_code`	Examples: 18K, 21K, 22K, 24K
`gross_weight_value`	Total item weight including stones and components
`net_weight_value`	Net calculated weight
`metal_weight_value`	Actual metal-only weight used in gold pricing
`weight_uom`	Gram or another enabled weight unit
`hallmark_reference`	Hallmark or stamping information
`certificate_number`	Item or gemstone certificate reference

The SRS item dictionary already defines metal type, karat, gross weight, net weight, metal weight, stone quantity, total carat, certificate, hallmark, barcode, serial number, making charge, wastage, and selling-price information as part of the jewelry item model.

Validation Rules

- Weight values cannot be negative.
- A weighted gold item must contain a weight.
- Karat is required when the selected metal requires purity identification.
- Metal weight cannot normally exceed gross weight.
- Item code must be unique within the tenant or company.
- Duplicate serial numbers, barcodes, certificate numbers, and style codes must be detected.
- Required fields may change based on item type.

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4. Stones and Gemstones

The item form must support stone information.

At minimum, capture:

- Stone type.
- Stone name.
- Stone quantity.
- Total carat weight.
- Stone color.

- Stone clarity.
- Cut.
- Shape.
- Certificate number.
- Stone cost or value.
- Supplier or certificate attachment, when applicable.

For items containing multiple stone types, the recommended implementation is to create a related table such as:

```
item_stones
```

Each item may then have one or more stone records.

Suggested fields:

```
item_stone_id
company_id
item_id
stone_type_id
stone_name
quantity
carat_value
color
clarity
cut
shape
certificate_number
unit_cost_amount
total_cost_amount
notes
```

The stone total must be available to the pricing engine.

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5. Making Charges and Jewelry Cost Components

The item and pricing modules must support the following commercial components:

- Gold or metal value.
- Making charge.
- Making charge type.

- Wastage percentage or value.
- Stone value.
- Labor amount.
- Additional material cost.
- Markup.
- Tax.
- Discount.
- Final selling price.

Making Charge Types

The system should support configurable making-charge methods such as:

- Fixed amount per item.
- Amount per gram.
- Percentage of metal value.
- Manually entered amount.
- Formula-based amount.

The selected method and calculated amount must be visible before the sale is posted.

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6. Daily Gold Rate Module

A dedicated **Gold Rates** module must be added or activated.

The system shall support manually entered and externally sourced rates for gold, silver, platinum, diamond, or other configured references. Rates must be definable by metal, karat, currency, source, and effective timestamp.

Required Gold Rate Fields

```
gold_rate_id
company_id
branch_id
metal_type
karat_code
currency_code
rate_per_weight_unit
weight_uom
rate_source
```

```
effective_at
expires_at
status
created_by
created_at
updated_at
```

Required Behaviors

- Each tenant may maintain its own gold rates.
- Rates must be separated by karat.
- Rates must support different currencies.
- The latest valid rate should be selected automatically.
- Authorized users may enter or update the daily rate.
- Rate changes must be logged.
- Branch-specific rates may override company rates when enabled.
- Old rates must remain available for reporting and audit.
- Updating today’s rate must not change old posted invoices.

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7. Jewelry Pricing Engine

The team must implement a reusable jewelry pricing service. Pricing logic must not be written separately inside every screen.

The same pricing engine must be used by:

- Item form.
- Price preview.
- POS.
- Quotations.
- Sales orders.
- Invoices.
- Customer portal in later phases.
- APIs and integrations.

Recommended Calculation Sequence

```
Metal Value =
Metal Weight × Applicable Gold Rate

Wastage Value =
```

```
Configured Wastage Rule

Making Charge =
Fixed Amount, Amount per Gram, Percentage, or Manual Value

Base Jewelry Value =
Metal Value
+ Wastage Value
+ Making Charge
+ Stone Value
+ Labor
+ Additional Components

Selling Price =
Base Jewelry Value
+ Markup
+ Tax
- Discount
```

The exact calculation must remain configurable because different jewelry businesses may use different commercial rules.

The SRS requires formulas that can include metal value, making charge, wastage, stone value, labor, markup, and rounding rules, together with a price preview and a stored snapshot of the rate and calculation applied to posted documents.

Pricing Preview

Before saving an item price or posting a sale, display a clear breakdown:

```
Gold Rate
Metal Weight
Metal Value
Making Charge
Wastage
Stone Value
Labor
Markup
Discount
Tax
Final Price
```

Users must be able to understand how the final price was produced.

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8. Item Screens to Customize

The following Quantro screens must be updated.

8.1 Item Creation and Editing

Add organized sections:

1. General Information.
2. Metal and Karat.
3. Weight Information.
4. Stones and Certificates.
5. Cost and Making Charges.
6. Pricing Preview.
7. Inventory Location.
8. Images and Attachments.
9. Audit and History.

Do not place all fields in one long, unstructured form.

8.2 Item List

Add filters for:

- Metal type.
- Karat.
- Category.
- Gross-weight range.
- Metal-weight range.
- Stone type.
- Certificate number.
- Branch.
- Location.
- Available, reserved, sold, repair, or damaged status.
- Memo or consignment ownership.
- Price range.

8.3 Item Details

Display:

- Item images.
- Barcode and serial number.

- Current branch and location.
- Complete gold specification.
- Complete stone specification.
- Current calculated price.
- Rate used for calculation.
- Item movement history.
- Purchase and sales history.
- Certificates and attachments.

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9. Inventory Customization

Quantro inventory must support both:

- Quantity movements.
- Weight movements.

Each movement must record:

```
company
branch
location
item
movement type
quantity delta
weight delta
source document
user
date and time
```

Inventory movements must support receiving, sales, returns, exchanges, repairs, transfers, adjustments, production-related movements, and memo movements as modules are activated. The SRS also requires perpetual balances by company, branch, and location, with movement drill-down and negative-stock prevention.

For serialized jewelry, the system must track the individual piece.

For weighted or grouped inventory, the system must track both available quantity and available weight where applicable.

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10. POS and Invoice Customization

The existing Quantro POS must be updated so that jewelry items display:

- Item image.
- Barcode or serial number.
- Metal type.
- Karat.
- Gross weight.
- Metal weight.
- Stone information.
- Daily gold rate.
- Metal value.
- Making charge.
- Stone value.
- Discount.
- Tax.
- Final selling price.

The cashier must be able to preview the calculation before completing the transaction.

The SRS expects a barcode-ready POS, quotations, held and draft sales, invoices, multiple payment methods, branch-specific numbering, and controlled price overrides.

Posted Invoice Snapshot

When an invoice is posted, store:

- Gold rate used.
- Effective rate date.
- Metal weight used.
- Karat.
- Making charge.
- Wastage.
- Stone value.
- Pricing formula result.
- User who approved any override.

Posted invoices must not be recalculated when a new daily rate is entered.

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11. APIs and Backend Services

Use the current Quantro API structure and introduce or extend the following endpoints:

```
/api/v1/items
/api/v1/items/{id}
/api/v1/catalog/import
/api/v1/gold-rates
/api/v1/gold-rates/current
/api/v1/pricing/preview
/api/v1/inventory/balances
/api/v1/inventory/movements
/api/v1/invoices
/api/v1/payments
```

The SRS specifically identifies the item, inventory, sales, gold-rate, and pricing-preview API areas as part of the implementation direction.

All API operations must include:

- Tenant validation.
- Authorization validation.
- Server-side business validation.
- Consistent error responses.
- Audit information.
- Database transaction protection for posted operations.

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12. Recommended Development Order

The team shall proceed in the following order:

Step 1 — Review Quantro

Identify the current:

- Tenant model.
- Item tables.
- Inventory tables.

- POS workflow.
- Invoice workflow.
- Pricing logic.
- Permission system.
- APIs and frontend components.

Step 2 — Freeze Reference Dictionaries

Define:

- Metal types.
- Karats.
- Stone types.
- Weight units.
- Making-charge methods.
- Item statuses.
- Ownership types.
- Inventory movement types.

The SRS recommends freezing the business glossary, naming conventions, and reference dictionaries before migration or UI implementation begins.

Step 3 — Database and Migrations

- Extend the item schema.
- Create gold-rate tables.
- Create item-stone tables.
- Add pricing snapshot support.
- Add indexes and tenant foreign keys.
- Create seed data for standard karats and metal types.

Step 4 — Backend Services

- Item validation.
- Gold-rate service.
- Pricing engine.
- Inventory weight handling.
- Audit logging.
- Tenant scoping.

Step 5 — Frontend Customization

- Jewelry item form.
- Gold-rate management screen.
- Price-preview component.
- Item details and listing.
- Inventory screens.
- Jewelry-aware POS.
- Invoice and receipt templates.

Step 6 — Testing and UAT

Test every flow using multiple tenants, branches, locations, metals, karats, weights, and currencies.

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13. Minimum Acceptance Scenario

The first implementation shall not be accepted until the team can demonstrate the following complete scenario:

1. Create a new tenant or jewelry company.
2. Create a branch, showroom, safe, and warehouse location.
3. Configure gold as a metal type.
4. Configure 18K, 21K, 22K, and 24K.
5. Enter today's gold rate for 21K.
6. Create a 21K gold ring.
7. Enter gross weight, net weight, and metal weight.
8. Add stone quantity, carat, color, clarity, and stone value.
9. Add making charge and wastage.
10. Preview the complete calculated price.
11. Receive the item into inventory.
12. Transfer it from the safe to the showroom.
13. Scan the item in the POS.
14. Display the jewelry price breakdown.
15. Complete payment and issue an invoice.
16. Store the gold-rate and pricing snapshot on the invoice.
17. Reduce stock automatically.
18. Display the item movement history.
19. Verify that another tenant cannot access any of the previous data.

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14. Current Phase Exclusions

The team shall not delay the Retail Core by beginning the following advanced modules now:

- Casting.
- Bill of materials.
- Manufacturing work orders.
- Factory production.
- Salesman inventory.
- Wholesale memo.
- Advanced appraisal.
- Scrap and trade-in.
- Full customer portal.
- Dynamic online repricing.

The database and service architecture should remain ready for these future phases, but the immediate development priority is:

Jewelry Item Master + Daily Gold Rates + Pricing Engine + Weight-Based Inventory + Gold-Aware POS and Invoicing.

Final Instruction to the Development Team

The first goal is not to redesign every part of Quantro. The first goal is to make the existing Quantro Retail Core correctly understand and manage a jewelry item.

A jewelry item is not an ordinary product with only a name, quantity, and fixed price. It may have:

- A metal type.
- A karat.
- Multiple weight values.
- One or more stones.
- A changing daily gold rate.
- Making charges.
- Wastage.
- Certificates.
- A calculated selling price.

- A unique serialized stock identity.

All development decisions during the first implementation stage must support this model consistently across the **database, APIs, item screens, inventory, POS, invoices, reports, permissions, and tenant isolation**.